

Cloudflare Radar

2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's global network during 2024.

Audience: Technology Executives & Network Infrastructure Leaders

radar.cloudflare.com/year-in-review/2024

17%

Traffic Growth

225

Disruptions

6.5%

Mitigated

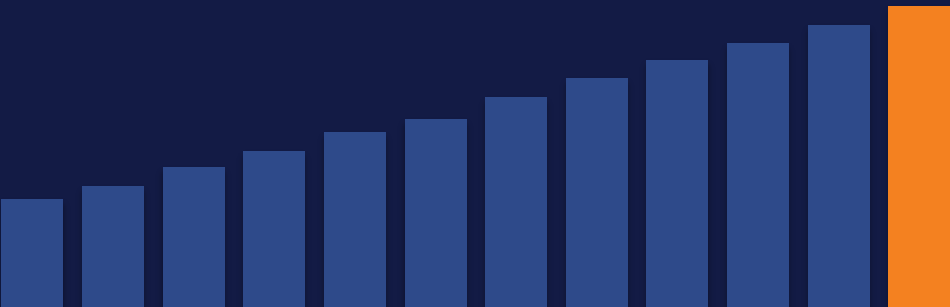
17% Global Traffic Growth and Starlink's 3.3x Surge

Internet traffic continues its upward trajectory | Satellite connectivity reaches underserved regions

Global Internet Traffic Growth

17%

Year-over-Year Growth in 2024

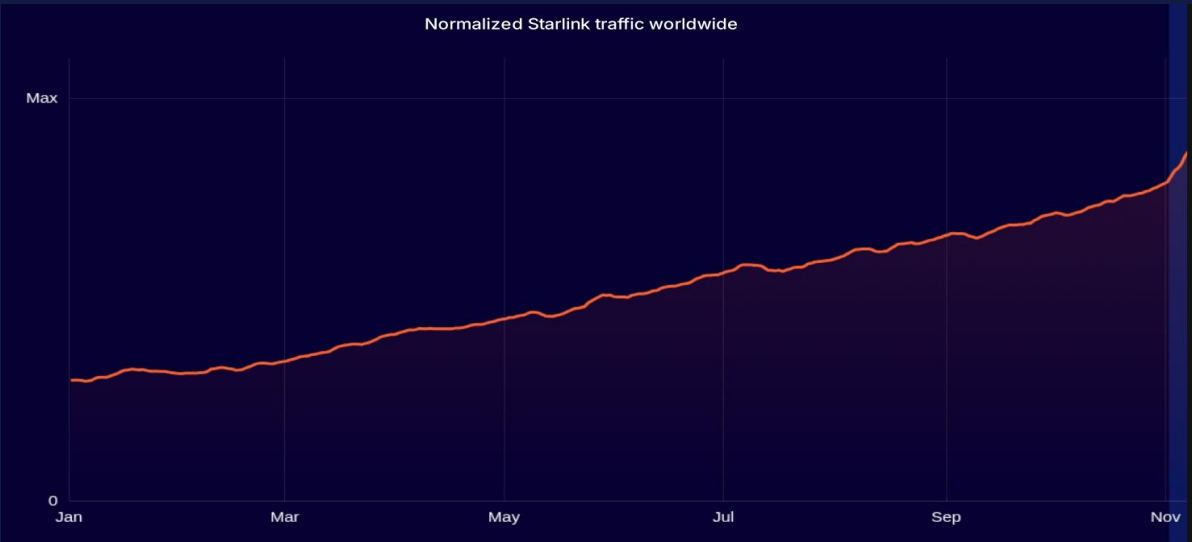


Jan Dec
Normalized monthly traffic index, 2024

Starlink Traffic Surge

3.3x

Growth in SpaceX Starlink Traffic (2024)



Satellite connectivity expanding to previously underserved locations globally.

Google Leads Top 10 Services as Generative AI Rises

Rankings based on Cloudflare 1.1.1.1 DNS resolver aggregate data | 2nd consecutive year for Google at #1

Rank	Service	Category
#1	Google	Search, Cloud, Advertising
#2	Facebook	Social Media
#3	Apple	Consumer Tech, App Store
#4	TikTok	Short-form Video
#5	AWS	Cloud Infrastructure
#6	Microsoft	Enterprise, Cloud, OS
#7	Instagram	Social / Photo Sharing
#8	YouTube	Video Streaming
#9	Amazon	E-commerce, Cloud
#10	WhatsApp	Messaging

Generative AI: The Defining Trend

Generative AI services saw explosive growth throughout 2024, rising rapidly across consumer and enterprise segments.

The Metaverse, by contrast, receded significantly from its earlier prominence.

AI-first products are reshaping DNS query patterns and Internet traffic distribution globally.

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

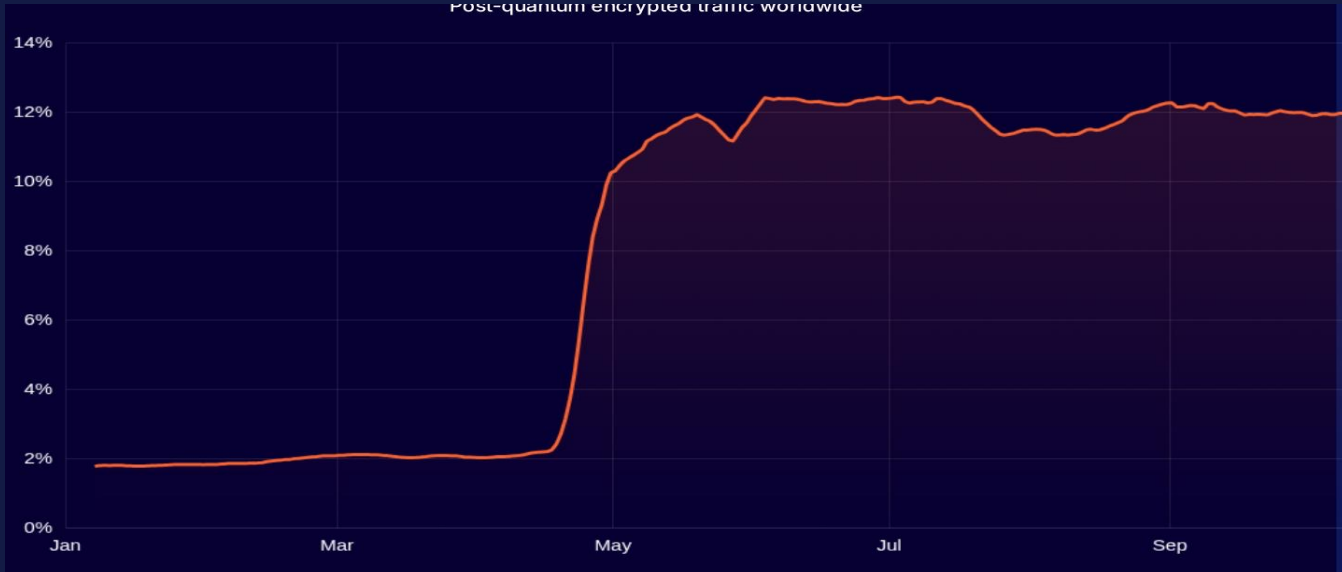
Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Chrome 124 (Apr 17) triggered rapid adoption | HTTP/3 at 21% | Go leads API clients

Post-Quantum Encryption Adoption

% of TLS 1.3 traffic using post-quantum key agreement

Post-quantum encrypted traffic worldwide



Chrome 124 default → April 17, 2024

Mozilla Firefox rolling out PQC by default. Apple Safari in initial testing phase.

Additional Context

Mobile Traffic	iOS Share	IPv6	Search Engine
41% from mobile devices globally	33% of traffic from iOS	28% dual-stack adoption	Google 89% of search referrals

HTTP/3 Adoption

21%

of traffic uses HTTP/3 over QUIC
Faster connections, loss mitigation, encryption by default

Post-Quantum Encryption

2%→13%

of TLS 1.3 traffic
Jumped after Chrome 124 enabled PQC key agreement by default

Top API Client Language

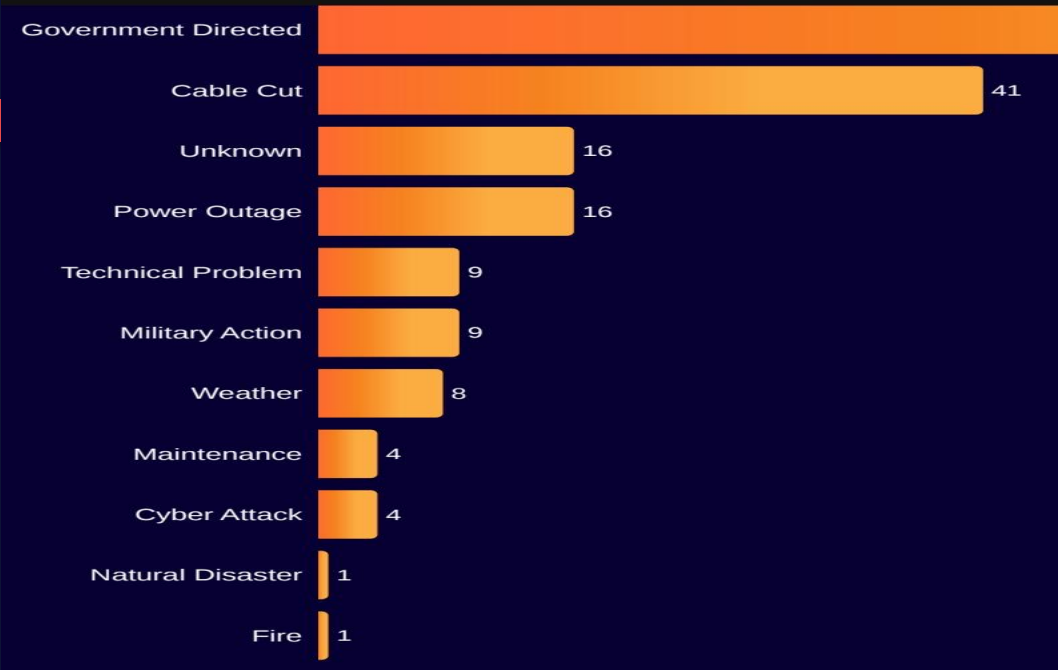
Go

~12% of automated API requests ahead of Node.js, Python (9.6%), Java (7.4%), .NET (3.6%)

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Cloudflare Radar Outage Center | Tracked by scope and duration using Cloudflare traffic data

Cause	Count	Share
Government Directed	116+	
Cable Cut	41	
Unknown	16	
Power Outage	16	
Technical Problem	9	
Military Action	9	
Weather	8	
Maintenance	4	
Cyber Attack	4	
Natural Disaster	1	
Fire	1	
Total: 225 major disruptions observed globally in 2024		



Key Insights

- Government-directed shutdowns accounted for 116+ incidents—more than all other causes combined.
- Cable cuts (41) represent physical infrastructure fragility, often affecting multiple countries simultaneously.
- Power outages (16) and technical problems (9) highlight operational resilience gaps.
- Only 4 disruptions were directly caused by cyber attacks, though their impact can be disproportionate.

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Cloudflare security data | Bot traffic, vulnerability exploitation, email threats

6.5%

of all traffic mitigated

68%

of global bot traffic from top 10 countries

4.3%

of emails processed were malicious

43%

of malicious emails contained deceptive links

Most-Attacked Sector

 **Gambling & Games**

Highest attack volume across all industry verticals tracked by Cloudflare WAF and DDoS mitigation systems in 2024.

68% of bot traffic originates from just the top 10 countries, highlighting geographic concentration of attack infrastructure.

Log4j Persistence

CVE-2021-44228

First disclosed: December 2021

Still actively exploited in 2024—three+ years after public disclosure.

Demonstrates the long tail of critical vulnerability exploitation.

Unpatched systems remain exposed years after patches are available.

Email & Routing Security

4.3%

of all emails were malicious

43%

of malicious emails had deceptive links

.bar

TLD with highest malicious/spam email share

+6.4%

increase in RPKI valid routes

Five Takeaways Shaping the 2025 Internet Outlook

Strategic implications for network and security leaders

01

Internet Traffic Growth Continues

Global traffic grew 17% in 2024, driven by mobile, video, and AI services. Starlink's 3.3x surge signals satellite broadband is becoming a mainstream connectivity layer. Plan capacity and CDN strategy for continued double-digit growth.

02

Post-Quantum Encryption Is Here Now

Adoption leapt from 2% to 13% of TLS 1.3 traffic after Chrome 124's April 2024 default enablement. Firefox and Safari are following. Begin auditing TLS infrastructure for PQC readiness.

03

Connectivity Fragility from Government Actions

116+ of 225 disruptions were government-directed—more than all other causes combined. Multi-path routing, anycast, and resilient DNS strategies are essential for global operations.

04

Persistent Threats Demand Continuous Vigilance

6.5% of all traffic mitigated; Log4j still exploited 3+ years on. Gambling/Games most targeted. 4.3% of emails malicious. Assume baseline attack pressure is permanent, not episodic.

05

Developer Ecosystem Reshaping Traffic Patterns

Go now leads API client languages (~12%). Generative AI services surged while Metaverse faded. Track developer platform shifts to anticipate traffic architecture and security surface changes.